CISS350: Data Structures and Advanced Algorithms
Quiz q10701

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You are given the following binary tree node class:

```
class BTNode
{
public:
    int key_;
    BTNode * parent_, * left_, * right_;
};
```

The following questions uses this class.

(You should create test cases to check that your functions work.)

Q1. Complete the function below that returns the pointer pointing to the sibling node. For instance of a node `n1` has left child `n2` and right child `n3`, and `p` points to node `n2`, then `sibling(p)` will return the address of `n3`. If there is no sibling, `NULL` is returned. (For instance if `p` points to the root node, `NULL` is returned. Also, if the parent of `*p` has only one child, then `NULL` is also returned.)
Answer:

```
BTNode * sibling(BTNode * p)
{
  if (p == NULL)
      return NULL;

  BTNode * q = p->parent_;

  if(q->left_ == p)
      return q->right_;
  if(q->right_ == p)
      return q->left_;
  else
      return NULL;
}
```

Q2. Complete the following function that returns the size of the tree that `p` points to (i.e., `p` points to the root of this tree).
Answer:

```
int size(BTNode * p)
{
  if (p == NULL)
      return 0;

  BTNode * left = p->left_;
  BTNode * right = p->right_;

  return 1 + (size(left) + size(right));
}
```

Q3. Complete the following function that returns the maximum value of the tree that `p` points to (i.e., `p` points to the root of this tree). In other words, it returns the maximum of all the `key_` values of the nodes in the tree that `p` points to.
Answer:

```
int max(BTNode * p)
{
  if(p == NULL)
      return 0;

  int leftMax = max(p->left_);
  int rightMax = max(p->right_);

  int m = p->key_;

  if(leftMax > m) m = leftMax;
  if(rightMax > m) m = rightMax;

  return m;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.